



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2021 - 2022)

JEE (Main)
FULL SYLLABUS
28-03-2022

JEE(Main+Advanced) : ENTHUSIAST COURSE : [SCORE (ALL STAR BATCH)]

ANSWER KEY

PART-1 : PHYSICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	D	B	C	B	D	D	B	B	B	C
	Q.	11	12	13	14	15	16	17	18	19	20
SECTION-II	A.	B	D	B	B	B	D	A	A	C	D
	Q.	1	2	3	4	5	6	7	8	9	10
SECTION-II	A.	25	13	74	80	10	22	6000	50	2	10

PART-2 : CHEMISTRY

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	A	A	C	A	B	C	D	C	C	D
	Q.	11	12	13	14	15	16	17	18	19	20
SECTION-II	A.	B	C	C	A	B	C	D	A	D	A
	Q.	1	2	3	4	5	6	7	8	9	10
SECTION-II	A.	10	0	2	25	25	24	101	26	18	464

PART-3 : MATHEMATICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	B	D	A	A	A	A	C	C	D	D
	Q.	11	12	13	14	15	16	17	18	19	20
SECTION-II	A.	D	D	A	A	A	A	B	A	C	C
	Q.	1	2	3	4	5	6	7	8	9	10
SECTION-II	A.	4	3	164	3	1	16	238	1	210	11

HINT – SHEET

PART-1 : PHYSICS

SECTION-I

1. Ans (D)

$C_p - C_v = R$ for ideal gas and gas behaves as ideal gas at high temperature
so $T_p > T_Q$

2. Ans (B)

$I_z = I_x + I_y$ (using perpendicular axis theorem)
& $I = mk^2$ (K : radius of gyration)
so $mK_z^2 = mK_x^2 + mK_y^2$
 $K_z^2 = K_x^2 + K_y^2$
so radius of gyration about axes x, y & z won't be same hence assertion A is not correct reason
R is correct statement (property of a rigid body)

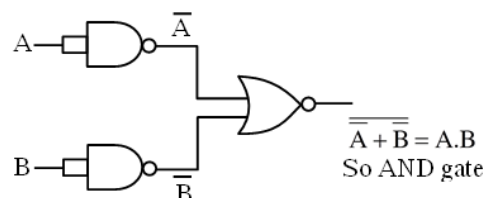
3. Ans (C)

$$\lambda = \frac{h}{p} = \frac{h}{\sqrt{2mE}} \propto \frac{1}{\sqrt{m}}$$

$$m_\alpha > m_p > m_e$$

$$\text{so } \lambda_e > \lambda_p > \lambda_\alpha$$

4. Ans (B)



5. Ans (D)

$$\lambda = \frac{h}{p}$$

both the particles will move with momentum same in magnitude & opposite in direction.
So De-Broglie wavelength of both will be same
i.e. ratio 1 : 1

6. Ans (D)

$$\frac{3N_0}{4} = N_0 e^{-\lambda t_1}$$

$$\frac{N_0}{2} = N_0 e^{-\lambda t_2}$$

$$\ln(3/4) = -\lambda t_1 \dots (i)$$

$$\ln(1/2) = -\lambda t_2 \dots (ii)$$

$$\ln(3/4) - \ln(1/2) = \lambda(t_2 - t_1) \dots (iii)$$

$$\Delta t = \frac{\ln(3/2)}{\lambda}$$

7. Ans (B)

$$(a) \vec{C} = \vec{A} + \vec{B}$$

Option (iv)

$$(b) \vec{A} = \vec{B} + \vec{C} = \vec{C} + \vec{B}$$

Option (iii)

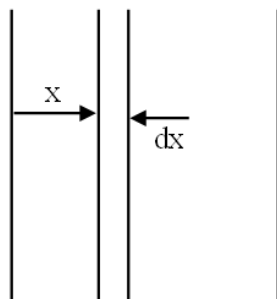
$$(c) \vec{B} = \vec{A} + \vec{C}$$

Option (i)

$$(d) \vec{A} + \vec{B} + \vec{C} = 0$$

Option (ii)

8. Ans (B)



Taking an element of width dx at a distance x ($x < d/2$) from left plate

$$dc = \frac{(\epsilon_0 + kx)A}{dx}$$

Capacitance of half of the capacitor

$$\frac{1}{C} = \int_0^{d/2} \frac{1}{dc} = \frac{1}{A} \int_0^{d/2} \frac{dx}{\epsilon_0 + kx}$$

$$\frac{1}{C} = \frac{1}{kA} \ln \left(\frac{\epsilon_0 + kd/2}{\epsilon_0} \right)$$

Capacitance of second half will be same

$$C_{eq} = \frac{C}{2} = \frac{kA}{2 \ln \left(\frac{2\epsilon_0 + kd}{2\epsilon_0} \right)}$$

9. Ans (B)

$$PV^r = \text{const.}$$

$$TV^{r-1} = \text{const.} T \left(\ell \right)^{\frac{5}{3}-1} = \text{const.}$$

$$\frac{T_1}{T_2} = \left(\frac{\ell_2}{\ell_1} \right)^{2/3}$$

10. Ans (C)

$$\sin \theta_C = \frac{1}{\mu} = \frac{1}{2\mu_2} < \sin \theta_C$$

$$\sin \theta > \sin \theta_C$$

$$\theta > \theta_C$$

Total internal reflection will happen

11. Ans (B)

In series combination $\Delta \ell = \ell_1 + \ell_2$

$$Y = \frac{F/A}{\Delta \ell / \ell} \Rightarrow \Delta \ell = \frac{F \ell}{AY}$$

$$\Rightarrow \Delta \ell \propto \frac{\ell}{Y}$$

Equivalent length of rod after joining is $= 2\ell$

As, lengths are same and force is also same in series

$$\Delta \ell = \Delta \ell_1 + \Delta \ell_2$$

$$\frac{\ell_{eq}}{Y_{eq}} = \frac{\ell}{Y_1} + \frac{\ell}{Y_2} \Rightarrow \frac{2\ell}{Y} = \frac{\ell}{Y_1} + \frac{\ell}{Y_2}$$

$$\therefore Y = \frac{2Y_1 Y_2}{Y_1 + Y_2}$$

12. Ans (D)

$$A = \lambda N$$

$$\lambda = \frac{\ln 2}{t_{1/2}} = \frac{\ln 2}{3 \times 24 \times 60 \times 60} \text{sec}^{-1} = 2.67 \times 10^{-6} \text{sec}^{-1}$$

N = Number of atoms in 2 mg Au

$$= \frac{2 \times 10^{-3}}{198} \times 6 \times 10^{23} = 6.06 \times 10^{15}$$

$$A = \lambda N = 1.618 \times 10^{13} = 16.18 \times 10^{12} \text{ dps}$$

13. Ans (B)

Impulse = change in momentum

$$\text{Ball (a)} \quad |\Delta \vec{p}| = 2mu = J_1$$

$$\text{Ball (b)} \quad |\Delta \vec{p}| = 2mu \cos 45^\circ = J_2$$

$$\frac{J_1}{J_2} = \frac{1}{\cos 45^\circ} = \sqrt{2}$$

14. Ans (B)

$$\text{Fringe Width, } \beta = \frac{\lambda D}{d}$$

$$\beta_{\max} \Rightarrow d_{\min} \text{ and } \beta_{\min} \Rightarrow d_{\max}$$

$$d = d_0 + a_0 \sin \omega t$$

$$d_{\max} = d_0 + a_0 \text{ and } d_{\min} = d_0 - a_0$$

$$\therefore \beta_{\min} = \frac{\lambda D}{d_0 + a_0} \text{ and } \therefore \beta_{\max} = \frac{\lambda D}{d_0 - a_0}$$

$$\beta_{\max} - \beta_{\min} = \frac{\lambda D}{d_0 - a_0} - \frac{\lambda D}{d_0 + a_0} = \frac{2\lambda D a_0}{d_0^2 - a_0^2}$$

15. Ans (B)

$$\left(\frac{\Delta Q}{\Delta t} \right)_A = \left(\frac{\Delta Q}{\Delta t} \right)_B$$

$$m S_A \left(\frac{\Delta T}{\Delta t} \right)_A = m S_B \left(\frac{\Delta T}{\Delta t} \right)_B$$

$$\frac{S_A}{S_B} = \frac{\left(\frac{\Delta T}{\Delta t} \right)_A}{\left(\frac{\Delta T}{\Delta t} \right)_B} = \frac{90/6}{120/3} = \frac{15}{40} = \frac{3}{8}$$

16. Ans (D)

Reflected wave will have direction opposite to incident wave.

17. Ans (A)

Max. voltage that can be measured by this potentiometer will be equal to potential drop across AB

$$R_{AB} = 10 \times 0.1 \times 100 = 100 \text{ ohm.}$$

$$\therefore V_{AB} = \frac{6}{20 + 100} \times 100 = 6 \times \frac{100}{120} = 5V$$

18. Ans (A)

$$\text{Bandwidth} = 2 \times f_m$$

$$= 2 \times 10^5 \text{ HZ} = 200 \text{ KHZ}$$

19. Ans (C)

In 4 sec. 1st drop will travel

$$\Rightarrow \frac{1}{2} \times (9.8) \times (4)^2 = 78.4 \text{ m}$$

$\therefore$ 2nd drop would have travelled

$$\Rightarrow 78.4 - 34.3 = 44.1 \text{ m. Time for 2nd drop}$$

$$\frac{1}{2} (9.8) t^2 = 44.1$$

$$\boxed{t = 3 \text{ sec}}$$

$\therefore$ each drop have time gap of 1 sec

$\therefore$ 1 drop per sec

20. Ans (D)

Angular momentum conservation equation

$$v_0 x_2 = v_1 x_1$$

$$v_1 = \frac{v_0 x_2}{x_1}$$

PART-1 : PHYSICS

SECTION-II

1. Ans (25)

$$p_i = p_f$$

$$2 \times 4 = 2 \times 1 + m_2 \times v_2$$

$$m_2 v_2 = 6 \dots (i)$$

by coefficient of restitution

$$1 = \frac{v_2 - 1}{4} \Rightarrow v_2 = 5 \text{ m/s}$$

by (i)

$$m_2 \times 5 = 6$$

$$m_2 = 1.2 \text{ kg}$$

$$v_{cm} = \frac{m_1 v_1 + m_2 v_2}{m_1 + m_2}$$

$$v_{cm} = \frac{2 \times 1 + 1.2 \times 5}{2 + 1.2} = \frac{8}{3.2} = \frac{25}{10}$$

$$\boxed{x = 25}$$

2. Ans (13)

For (A)

$$\text{Reading} = \text{MSR} + \text{CSR} + \text{Error}$$

$$0.322 = 0.300 + \text{CSR} + 5 \times \text{LC}$$

$$0.322 = 0.300 + \text{CSR} + 0.005$$

$$\text{CSR} = 0.017$$

For B

$$\text{Reading} = \text{MSR} + \text{CSR} + \text{Error}$$

$$0.322 = 0.200 + \text{CSR} + 0.092$$

$$\text{CSR} = 0.030$$

$$\text{Difference} = 0.030 - 0.017 = 0.013 \text{ cm}$$

$$\text{Division on circular scale} = \frac{0.013}{0.001} = 13$$

3. Ans (74)

$$I_{\max} = \frac{V}{R} = \frac{20V}{10K\Omega} = 2mA$$

For LR – decay circuit

$$I = I_{\max} e^{-Rt/L}$$

$$I = 2mA e^{\frac{-10 \times 10^3 \times 1 \times 10^{-6}}{10 \times 10^{-3}}}$$

$$I = 2mA e^{-1}$$

$$I = 2 \times 0.37 mA$$

$$I = \frac{74}{100} mA$$

$$x = 74$$

4. Ans (80)

$$\phi = \vec{B} \cdot \vec{S}$$

$$\phi = \frac{4}{\pi} \times 10^{-3} \left(1 - \frac{t}{100} \right) \cdot \pi R^2$$

$$\phi = 4 \times 10^{-3} \times (1)^2 \left(1 - \frac{t}{100} \right)$$

$$\varepsilon = \frac{-d\phi}{dt}$$

$$\varepsilon = \frac{-d}{dt} \left(4 \times 10^{-3} \left(1 - \frac{t}{100} \right) \right)$$

$$\varepsilon = 4 \times 10^{-3} \left(\frac{1}{100} \right) = 4 \times 10^{-5} V$$

When $B = 0$

$$1 - \frac{t}{100} = 0$$

$$t = 100 \text{ sec}$$

$$\text{Heat} = \frac{\varepsilon^2}{R} t$$

$$\text{Heat} = \frac{(4 \times 10^{-5})^2}{2 \times 10^{-6}} \times 100 J$$

$$\text{Heat} = \frac{16 \times 10^{-10} \times 100}{2 \times 10^{-6}} J$$

$$\text{Heat} = 0.08 J$$

$$\text{Heat} = 80 mJ$$

5. Ans (10)

$$\omega = \sqrt{\frac{k_{eq}}{\mu}}$$

μ = reduced mass

springs are in series connection

$$k_{eq} = \frac{k_1 k_2}{k_1 + k_2}$$

$$k_{eq} = \frac{k \times 4k}{5k} = \frac{4k}{5}$$

$$k_{eq} = \frac{4 \times 20}{5} N/m = 16 N/m$$

$$\mu = \frac{m_1 m_2}{m_1 + m_2} = \frac{0.2 \times 0.8}{0.2 + 0.8} = 0.16 kg$$

$$\omega = \sqrt{\frac{16}{0.16}} = \sqrt{100} = 10$$

6. Ans (22)

$$B = \mu_0 (H + I)$$

$$B = \mu_0 H \left(1 + \frac{I}{H} \right)$$

$$B = B_0 (1 + x)$$

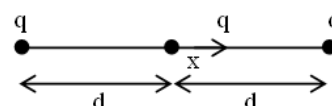
$$B - B_0 = B_0 x$$

$$\frac{B - B_0}{B_0} = x$$

$$\frac{B - B_0}{B_0} \times 100 = 100x$$

$$= 2.2 \times 10^{-3} = \frac{22}{10^4}$$

7. Ans (6000)



Net force on free charged particle

$$F = \frac{kq^2}{(d+x)^2} - \frac{kq^2}{(d-x)^2}$$

$$F = -kq^2 \left[\frac{4dx}{(d^2 - x^2)^2} \right]$$

$$a = -\frac{4kq^2 d}{m} \left(\frac{x}{d^4} \right)$$

$$a = -\left(\frac{4kq^2}{md^3} \right) x$$

So, angular frequency

$$\omega = \sqrt{\frac{4kq^2}{md^3}}$$

$$\omega = \sqrt{\frac{4 \times 9 \times 10^9 \times 10}{1 \times 10^{-6} \times 1^3}}$$

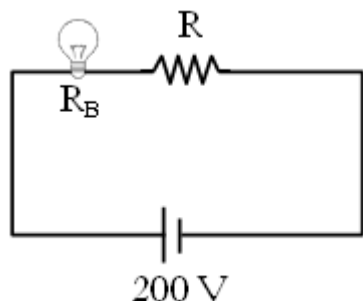
$$\omega = 6 \times 10^8 \text{ rad/sec}$$

8. Ans (50)

$$\text{Power, } P = \frac{V^2}{R_B}$$

$$R_B = \frac{V^2}{P} = \frac{100 \times 100}{200}$$

$$R_B = 50 \Omega$$



To produce same power, same voltage (i.e. 100 V) should be across the bulb

$$\text{Hence, } R = R_B$$

$$R = 50 \Omega$$

9. Ans (2)

Applying work energy theorem :

$$w_g + w_T = \Delta K$$

$$-mgl(1-\cos 60^\circ) = \frac{1}{2}mv^2 - \frac{1}{2}mu^2$$

$$v^2 = u^2 - 2gl(1-\cos 60^\circ)$$

$$v^2 = 9 - 2 \times 10 \times 0.5 \left(\frac{1}{2} \right)$$

$$v^2 = 4$$

$$v = 2 \text{ m/s}$$

10. Ans (10)

$$\vec{r} = 10\alpha t^2 \hat{i} + 5\beta(t-5)\hat{j}$$

$$\vec{v} = 20\alpha t \hat{i} + 5\beta \hat{j}$$

$$\vec{L} = m(\vec{r} \times \vec{v})$$

$$= m[10\alpha t^2 \hat{i} + 5\beta(t-5)\hat{j}] \times [20\alpha t \hat{i} + 5\beta \hat{j}]$$

$$\vec{L} = m[50\alpha\beta t^2 \hat{k} - 100\alpha\beta(t^2 - 5t)\hat{k}]$$

$$\text{At } t=0, \vec{L} = \vec{0}$$

$$50\alpha\beta t^2 - 100\alpha\beta(t^2 - 5t) = 0$$

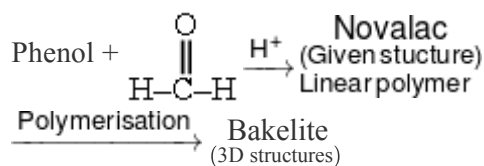
$$t - 2(t-5) = 0$$

$$t = 10 \text{ sec}$$

PART-2 : CHEMISTRY

SECTION-I

1. Ans (A)



2. Ans (A)

$$1. [\text{Fe}(\text{H}_2\text{O})_6]^{3+}$$

$$\text{Fe}^{3+} : [\text{Ar}]3d^5$$

$$\text{Hybridisation} : sp^3d^2$$

Magnetic nature : Paramagnetic (so this complex response to external magnetic field)

$$2. [\text{Ni}(\text{CN})_4]^{2-}$$

$$\text{Ni}^{2+} : [\text{Ar}]3d^8$$

$$\text{Hybridisation} : dsp^2$$

Magnetic nature : diamagnetic

$$3. [\text{Co}(\text{CN})_6]^{3-}$$

$$\text{Co}^{3+} : [\text{Ar}]3d^6$$

$$\text{Hybridisation} : d^2sp^3$$

Magnetic nature : diamagnetic

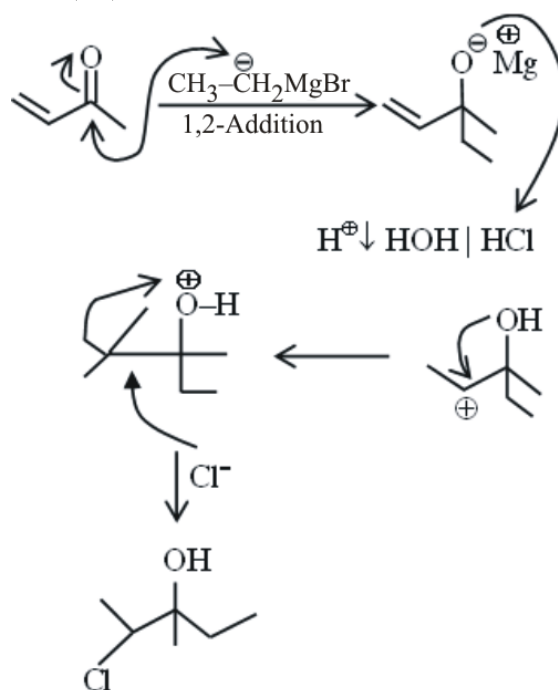
$$4. [\text{Ni}(\text{CO})_4]$$

$$\text{Ni} : [\text{Ar}] 3d^8 4s^2$$

$$\text{Hybridisation} : sp^3$$

Magnetic nature : diamagnetic

3. Ans (C)



4. **Ans (A)**

Forms spherical micelles with $\text{CH}_3(\text{CH}_2)_{16}$ group pointing towards the centre of sphere

5. **Ans (B)**

Albumin is water soluble.

6. **Ans (C)**

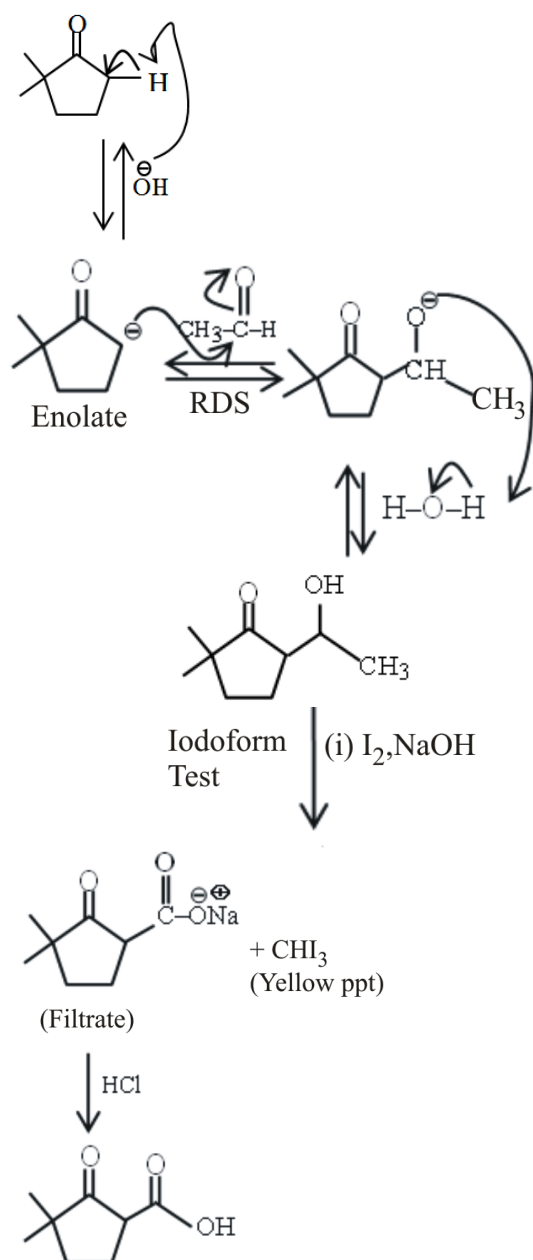
Enthalpy of bond dissociation (kJ/mole) at 298.2K

For , hydrogen = 435.88

For , Deuterium = 443.35

$\therefore E_H \simeq E_D - 7.5$

7. **Ans (D)**

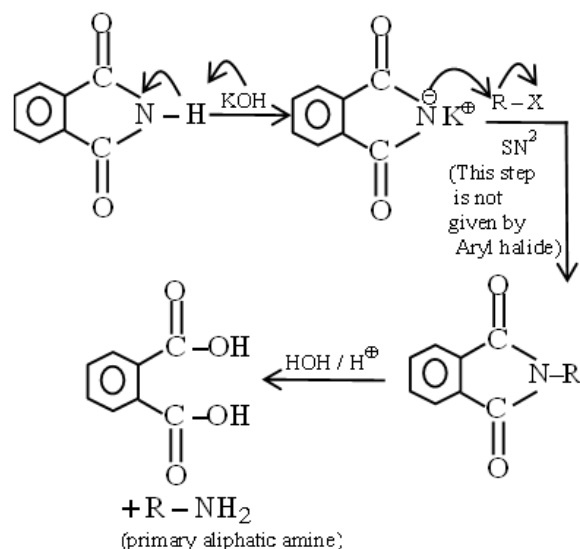


8. **Ans (C)**

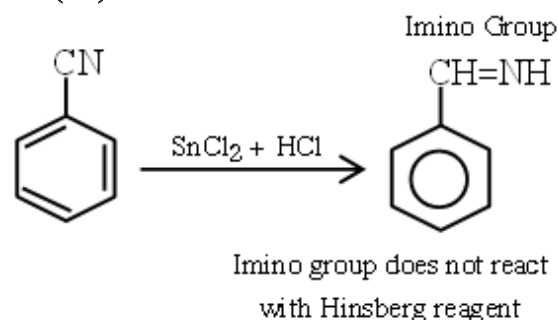
Bromine water gives tribromo products, other gives monobromo products in which para is major product.

9. **Ans (C)**

Gabriel phthalamide synthesis



11. **Ans (B)**



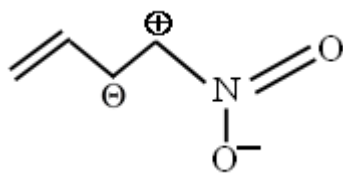
12. **Ans (C)**

Al^{3+} , Mg^{2+} and Na^+ are isoelectronic ionic species. For monoatomic ionic isoelectronic species as positive charge increases ionic size decreases. The order of size of Na^+ & K^+ is $\text{Na}^+ < \text{K}^+$,
 $\therefore$ order of ionic radii is : $\text{Al}^{3+} < \text{Mg}^{2+} < \text{Na}^+ < \text{K}^+$

13. **Ans (C)**

$[\text{SiCl}_6]^{2-}$ does not exist due to steric crowding of surrounding atoms.

14. Ans (A)



It is unstable RS (due to similar charge on adjacent atom)

15. Ans (B)

Statement-I is incorrect

$\text{Be}(\text{OH})_2$ dissolve in alkali due to its amphoteric nature.

Statement-II is correct

Solubility of alkaline earth metal hydroxide in water increases down the group due to rapid decreases in lattice energy as compared to hydration energy.

16. Ans (C)

(a) $\text{Cr}^{+6} \text{O}_3$ (b) $\text{Fe}^{+3} \text{O}_3$ (c) $\text{Mn}^{+4} \text{O}_2$

(d) $\text{V}^{+5} \text{O}_5$ (e) $\text{Cu}^{+1} \text{O}$

So order of oxidation state

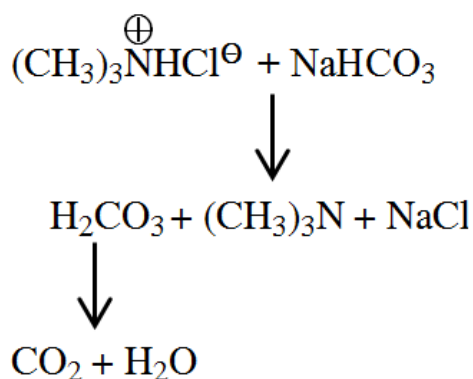
$a > d > c > b > e$

17. Ans (D)

CO_2 , CCl_4 and $\text{Cl}_2\text{C} = \text{CCl}_2$ are used as dry cleaning agents for clothes.

H_2O_2 is used as bleaching agent in laundry.

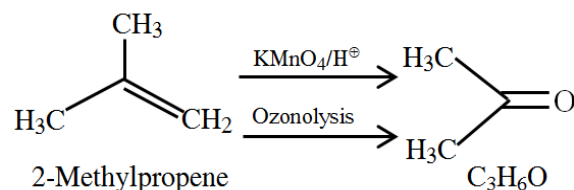
18. Ans (A)



19. Ans (D)

In bauxite impurities of Fe_2O_3 , TiO_2 and SiO_2 are present, Fe_2O_3 and TiO_2 are basic oxides therefore does not reacts with or dissolve in NaOH whereas SiO_2 is acidic oxide it gets dissolve in NaOH , hence leach out
 $\text{SiO}_2 + 2\text{NaOH} \rightarrow \text{Na}_2\text{SiO}_3(\text{aq.}) + \text{H}_2\text{O}$

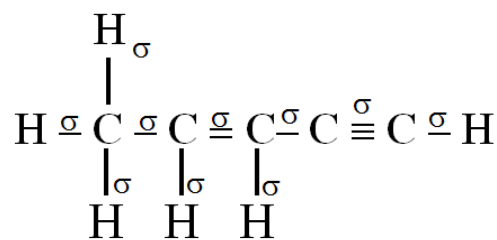
20. Ans (A)



PART-2 : CHEMISTRY

SECTION-II

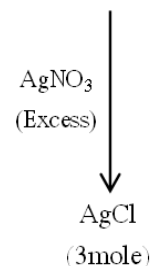
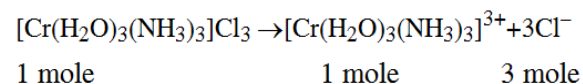
1. Ans (10)



numbers of s bonds = 10

2. Ans (0)

Mole of AgCl precipitated is equal the mole of Cl^- present in ionization sphere.



Since none of Cl^- is present in the co-ordination sphere. Therefore answer is zero.

3. **Ans (2)**

Total energy provided by

$$\text{Source per second} = \frac{1000}{10} = 100\text{J}$$

$$\text{Energy required to eject electron} = \frac{hc}{\lambda}$$

$$= \frac{6.626 \times 10^{-34}}{400 \times 10^{-9}} \times 3 \times 10^8$$

Number of electrons ejected

$$\begin{aligned} &= \frac{100}{\frac{6.626 \times 10^{-34} \times 3 \times 10^8}{400 \times 10^{-9}}} \\ &= \frac{400 \times 10^{-7} \times 10^{26}}{6.626 \times 3} \\ &= \frac{40 \times 10^{-20}}{6.626 \times 3} \\ &= 2.01 \times 10^{20} \end{aligned}$$

4. **Ans (25)**

From Henry's law

$$P_{\text{gas}} = K_H \cdot X_{\text{gas}}$$

$$0.835 = 1.67 \times 10^3 \times \frac{n(\text{CO}_2)}{\frac{0.9 \times 1000}{18}}$$

$$n(\text{CO}_2) = 0.025$$

$$\text{Millimoles of CO}_2 = 0.025 \times 1000 = 25$$

5. **Ans (25)**



$$\begin{array}{ccc} 1 & 1 & 1 \\ -x & -x & 2x \end{array}$$

$$\begin{array}{ccc} 1-x & 1-x & 1+2x \end{array}$$

$$K = \frac{[\text{C}]_{\text{eq}}^2}{[\text{A}]_{\text{eq}}[\text{B}]_{\text{eq}}} = \frac{(1+2x)^2}{(1-x)(1-x)}$$

$$100 = \left(\frac{1+2x}{1-x} \right)^2$$

$$\left(\frac{1+2x}{1-x} \right) = 10$$

$$x = \frac{3}{4}$$

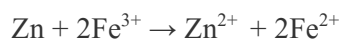
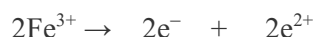
$$[\text{C}]_{\text{eq}} = 1 + 2x$$

$$= 1 + 2 \left(\frac{3}{4} \right)$$

$$= 2.5 \text{ M}$$

$$25 \times 10^{-1} \text{ M}$$

6. **Ans (24)**



$$E_{\text{cell}}^0 = 0.77 - (0.76) = 1.53 \text{ V}$$

$$1.50 = 1.53 - \frac{0.06}{2} \log \left(\frac{[\text{Fe}^{2+}]}{[\text{Fe}^{3+}]} \right)^2$$

$$\log \left(\frac{[\text{Fe}^{2+}]}{[\text{Fe}^{3+}]} \right) = \frac{0.03}{0.06} = \frac{1}{2}$$

$$\frac{[\text{Fe}^{2+}]}{[\text{Fe}^{3+}]} = 10^{1/2} = \sqrt{10}$$

$$\frac{[\text{Fe}^{3+}]}{[\text{Fe}^{2+}]} = \frac{1}{\sqrt{10}}$$

$$\frac{[\text{Fe}^{3+}]}{[\text{Fe}^{2+}] + [\text{Fe}^{3+}]} = \frac{1}{1 + \sqrt{10}} = \frac{1}{4.16}$$

$$= 0.2402$$

$$= 24 \times 10^{-2}$$

7. **Ans (101)**

$$\Delta H_{\text{sub}} = \Delta H_{\text{fus.}} + \Delta H_{\text{vap.}}$$

$$= 2.8 + 98.2$$

$$= 101 \text{ kJ/mol}$$

8. **Ans (26)**

$$n(\text{CH}_4) = \frac{PV}{RT}$$

$$= \frac{1 \times 4 \times 10^3 \times 1000}{0.083 \times 300}$$

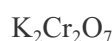
$$\text{Weight of CH}_4$$

$$= \frac{40 \times 16 \times 10^5}{0.083 \times 300} \text{ gm}$$

$$= 25.7 \times 10^5 \text{ gm}$$

9. **Ans (18)**

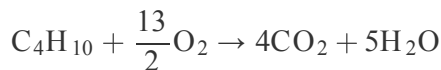
milli-equivalents of Fe^{2+} = milli-equivalents of



$$M \times 10 \times 1 = 0.02 \times 15 \times 6$$

$$M = 0.18 = 18 \times 10^{-2} \text{ M}$$

10. Ans (464)



$$\text{Moles of H}_2\text{O} = \frac{72}{18} = 4$$

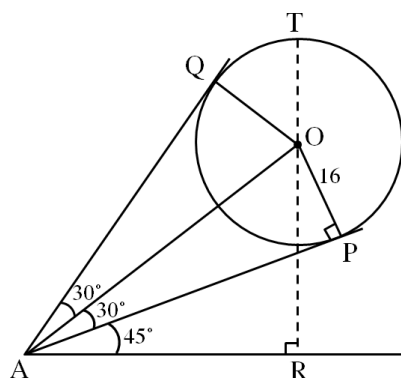
$$\text{Moles of C}_4\text{H}_{10} \text{ used} = \frac{1}{5} \times 4$$

$$\begin{aligned} \text{Weight of C}_4\text{H}_{10} \text{ used} &= \frac{4}{5} \times 58 \\ &= 46.4 \text{ gm} \end{aligned}$$

PART-3 : MATHEMATICS

SECTION-I

1. Ans (B)



O → centre of sphere

P, Q → point of contact of tangents from A

Let T be top most point of balloon & R be foot of perpendicular from O to ground.

From triangle OAP, $OA = 16 \csc 30^\circ = 32$

From triangle ABO, $OR = OA \sin 75^\circ$

$$= 32 \frac{(\sqrt{3} + 1)}{2\sqrt{2}}$$

So level of top most point = $OR + OT$

$$= 8(\sqrt{6} + \sqrt{2} + 2)$$

2. Ans (D)

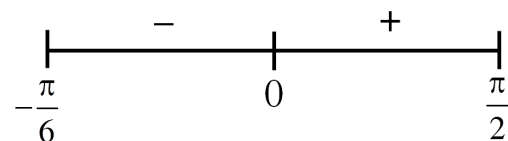
$$f(x) = 3\sin^4 x + 10\sin^3 x + 6\sin^2 x - 3,$$

$$x \in \left[-\frac{\pi}{6}, \frac{\pi}{2}\right]$$

$$f'(x) = 12\sin^3 x \cos x + 30\sin^2 x \cos x + 12\sin x \cos x$$

$$= 6\sin x \cos x (2\sin^2 x + 5\sin x + 2)$$

$$= 6\sin x \cos x (2\sin x + 1)(\sin x + 2)$$



Decreasing in $\left(-\frac{\pi}{6}, 0\right)$

3. Ans (A)

Let a be first term and d be common diff. of this A.P.

$$\text{Given } S_{3n} = 3S_{2n}$$

$$\Rightarrow \frac{3n}{2} [2a + (3n-1)d] = 3 \frac{2n}{2} [2a + (2n-1)d]$$

$$\Rightarrow 2a + (3n-1)d = 4a + (4n-2)d$$

$$\Rightarrow 2a + (n-1)d = 0$$

Now

$$\begin{aligned} \frac{S_{4n}}{S_{2n}} &= \frac{\frac{4n}{2} [2a + (4n-1)d]}{\frac{2n}{2} [2a + (2n-1)d]} = \frac{2 \left[\underbrace{2a + (n-1)d}_{=0} + 3nd \right]}{\left[\underbrace{2a + (n-1)d}_{=0} + nd \right]} \\ &= \frac{6nd}{nd} = 6 \end{aligned}$$

4. Ans (A)

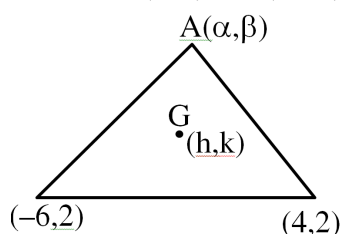
Given hyperbola is

$$16(x+1)^2 - 9(y-2)^2 = 164 + 16 - 36 = 144$$

$$\Rightarrow \frac{(x+1)^2}{9} - \frac{(y-2)^2}{16} = 1$$

$$\text{Eccentricity, } e = \sqrt{1 + \frac{16}{9}} = \frac{5}{3}$$

$$\Rightarrow \text{foci are } (4, 2) \text{ and } (-6, 2)$$



Let the centroid be (h, k)

& A(α, β) be point on hyperbola

$$\text{So } h = \frac{\alpha - 6 + 4}{3}, k = \frac{\beta + 2 + 2}{3}$$

$$\Rightarrow \alpha = 3h + 2, \beta = 3k - 4$$

(α, β) lies on hyperbola so

$$16(3h+2+1)^2 - 9(3k-4-2)^2 = 144$$

$$\Rightarrow 144(h+1)^2 - 81(k-2)^2 = 144$$

$$\Rightarrow 16(h^2 + 2h + 1) - 9(k^2 - 4k + 4) = 16$$

$$\Rightarrow 16x^2 - 9y^2 + 32x + 36y - 36 = 0$$

5. Ans (A)

If the vectors are co-planar,

$$\begin{vmatrix} a+b+2 & a+2b+c & -b-c \\ b+1 & 2b & -b \\ b+2 & 2b & 1-b \end{vmatrix} = 0$$

Now $R_3 \rightarrow R_3 - R_2$, $R_1 \rightarrow R_1 - R_2$

$$\text{So } \begin{vmatrix} a+1 & a+c & -c \\ b+1 & 2b & -b \\ 1 & 0 & 1 \end{vmatrix} = 0$$

$$\begin{aligned} &= (a+1)2b - (a+c)(2b+1) - c(-2b) \\ &= 2ab + 2b - 2ab - a - 2bc - c + 2bc \\ &= 2b - a - c = 0 \end{aligned}$$

6. Ans (A)

$$\lim_{x \rightarrow 2^+} f(x) = \lim_{x \rightarrow 2^+} e^{\frac{\tan(x-2)}{x-2}} = e^1$$

$$\lim_{x \rightarrow 2^-} f(x) = \lim_{x \rightarrow 2^-} \frac{-\lambda(x-2)(x-3)}{\mu(x-2)(x-3)} = -\frac{\lambda}{\mu}$$

$$\text{For continuity } \mu = e = -\frac{\lambda}{\mu} \Rightarrow \mu = e, \lambda = -e^2$$

$$\lambda + \mu = e(-e + 1)$$

7. Ans (C)

$$\text{Let } I = \int_{\pi/24}^{5\pi/24} \frac{(\cos 2x)^{1/3}}{(\cos 2x)^{1/3} + (\sin 2x)^{1/3}} dx \quad \dots(i)$$

$$\Rightarrow I = \int_{\pi/24}^{5\pi/24} \frac{(\cos\{2(\frac{\pi}{4}-x)\})^{1/3}}{(\cos\{2(\frac{\pi}{4}-x)\})^{1/3} + (\sin\{2(\frac{\pi}{4}-x)\})^{1/3}} dx$$

$$\left\{ \int_a^b f(x) dx = \int_a^b f(a+b-x) dx \right\}$$

$$\text{So } I = \int_{\pi/24}^{5\pi/24} \frac{(\sin 2x)^{1/3}}{(\sin 2x)^{1/3} + (\cos 2x)^{1/3}} dx \quad \dots(ii)$$

$$\text{Hence } 2I = \int_{\pi/24}^{5\pi/24} dx \quad [(i) + (ii)]$$

$$\Rightarrow 2I = \frac{4\pi}{24} \Rightarrow \boxed{I = \frac{\pi}{12}}$$

8. Ans (C)

$$\begin{aligned} &(a-b)^{-1} + (a-2b)^{-1} + \dots + (a-nb)^{-1} \\ &= \frac{1}{a} \sum_{r=1}^n \left(1 - \frac{rb}{a}\right)^{-1} \\ &= \frac{1}{a} \sum_{r=1}^n \left\{ \left(1 + \frac{rb}{a} + \frac{r^2 b^2}{a^2}\right) + (\text{terms to be neglected}) \right\} \\ &= \frac{1}{a} \left[n + \frac{n(n+1)}{2} \cdot \frac{b}{a} + \frac{n(n+1)(2n+1)}{6} \cdot \frac{b^2}{a^2} \right] \\ &= \frac{1}{a} \left[n^3 \left(\frac{b^2}{3a^2} \right) + \dots \right] \end{aligned}$$

$$\text{So } \gamma = \frac{b^2}{3a^3}$$

9. Ans (D)

$$\frac{dy - dx}{e^{y-x}} = x dx$$

$$\Rightarrow \frac{dy - dx}{e^{y-x}} = x dx$$

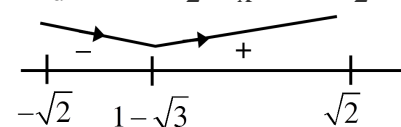
$$\Rightarrow -e^{x-y} = \frac{x^2}{2} + c$$

$$\text{At } x=0, y=0 \Rightarrow c=-1$$

$$\Rightarrow e^{x-y} = \frac{2-x^2}{2}$$

$$\Rightarrow y = x - \ln\left(\frac{2-x^2}{2}\right)$$

$$\Rightarrow \frac{dy}{dx} = 1 + \frac{2x}{2-x^2} = \frac{2+2x-x^2}{2-x^2}$$



So minimum value occurs at $x = 1 - \sqrt{3}$

$$y(1-\sqrt{3}) = (1-\sqrt{3}) - \ln\left(\frac{2-(4-2\sqrt{3})}{2}\right)$$

$$= (1-\sqrt{3}) - \ln(\sqrt{3}-1)$$

10. Ans (D)

$$(p \rightarrow q) \wedge (q \rightarrow \sim p)$$

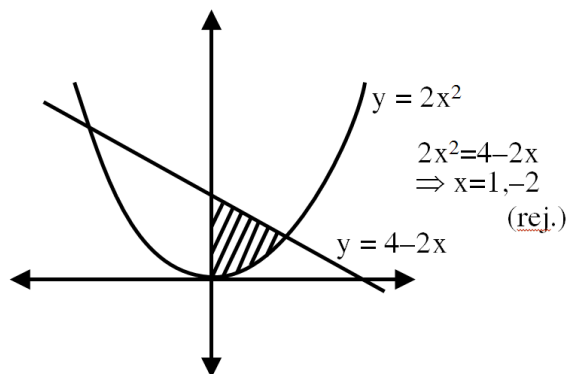
$$\equiv (\sim p \vee q) \wedge (\sim q \vee \sim p) \quad \{p \rightarrow q \equiv \sim p \vee q\}$$

$$\equiv (\sim p \vee q) \wedge (\sim p \vee \sim q) \quad \{\text{commutative property}\}$$

$$\equiv \sim p \vee (q \wedge \sim q) \quad \{\text{distributive property}\}$$

$$\equiv \sim p$$

11. Ans (D)



Required area

$$= \int_0^1 (4 - 2x - 2x^2) dx = 4x - x^2 - \frac{2x^3}{3} \Big|_0^1$$

$$= 4 - 1 - \frac{2}{3} = \frac{7}{3}$$

12. Ans (D)

$$(\sin x + \sin 4x) + (\sin 2x + \sin 3x) = 0$$

$$\Rightarrow 2 \sin \frac{5x}{2} \left\{ \cos \frac{3x}{2} + \cos \frac{x}{2} \right\} = 0$$

$$\Rightarrow 2 \sin \frac{5x}{2} \left\{ 2 \cos x \cos \frac{x}{2} \right\} = 0$$

$$2 \sin \frac{5x}{2} = 0 \Rightarrow \frac{5x}{2} = 0, \pi, 2\pi, 3\pi, 4\pi, 5\pi$$

$$\Rightarrow x = 0, \frac{2\pi}{5}, \frac{4\pi}{5}, \frac{6\pi}{5}, \frac{8\pi}{5}, 2\pi$$

$$\cos \frac{x}{2} = 0 \Rightarrow \frac{x}{2} = \frac{\pi}{2} \Rightarrow x = \pi$$

$$\cos x = 0 \Rightarrow x = \frac{\pi}{2}, \frac{3\pi}{2}$$

$$\text{So sum} = 6\pi + \pi + 2\pi = 9\pi$$

13. Ans (A)

$$g : \mathbb{N} \rightarrow \mathbb{N} \quad g(3n+1) = 3n+2$$

$$g(3n+2) = 3n+3$$

$$g(3n+3) = 3n+1$$

$$g(x) = \begin{cases} x+1 & x = 3k+1 \\ x+1 & x = 3k+2 \\ x-2 & x = 3k+3 \end{cases}$$

$$g(g(x)) = \begin{cases} x+2 & x = 3k+1 \\ x-1 & x = 3k+2 \\ x-1 & x = 3k+3 \end{cases}$$

$$g(g(g(x))) = \begin{cases} x & x = 3k+1 \\ x & x = 3k+2 \\ x & x = 3k+3 \end{cases}$$

If $f : \mathbb{N} \rightarrow \mathbb{N}$, f is a one-one function such that $f(g(x)) = f(x) \Rightarrow g(x) = x$, which is not the case

If $f : \mathbb{N} \rightarrow \mathbb{N}$, f is an onto function

such that $f(g(x)) = f(x)$,

one possibility is

$$f(x) = \begin{cases} n & x = 3n+1 \\ n & x = 3n+2 \\ n & x = 3n+3 \end{cases} \quad n \in \mathbb{N}_0$$

Here $f(x)$ is onto, also $f(g(x)) = f(x) \forall x \in \mathbb{N}$

14. Ans (A)

$$f : [0, \infty) \rightarrow [0, \infty), f(x) = \int_0^x [y] dy$$

Let $x = n + f$, $f \in (0, 1)$

$$\text{So } f(x) = 0 + 1 + 2 + \dots + (n-1) + \int_n^{n+f} n dy$$

$$f(x) = \frac{n(n-1)}{2} + nf$$

$$= \frac{[x]([x]-1)}{2} + [x] \{x\}$$

$$\text{Note } \lim_{x \rightarrow n^+} f(x) = \frac{n(n-1)}{2},$$

$$\lim_{x \rightarrow n^-} f(x) = \frac{(n-1)(n-2)}{2} + (n-1)$$

$$= \frac{n(n-1)}{2}$$

$$f(x) = \frac{n(n-1)}{2} \quad (n \in \mathbb{N}_0)$$

so $f(x)$ is cont. $\forall x \geq 0$ and diff. except at integer points

15. Ans (A)

$$D = \begin{vmatrix} 2 & 3 & 6 \\ 1 & 2 & a \\ 3 & 5 & 9 \end{vmatrix} = 3 - a$$

$$D = \begin{vmatrix} 2 & 3 & 8 \\ 1 & 2 & 5 \\ 3 & 5 & b \end{vmatrix} = b - 13$$

If $a = 3, b \neq 13$, no solution.

16. Ans (A)

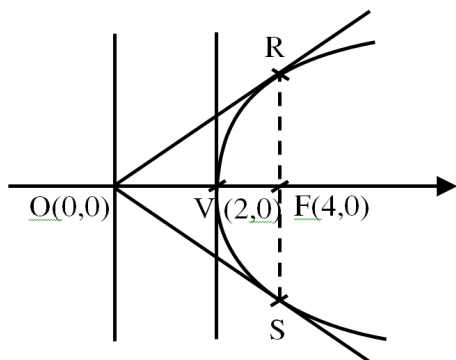
$$\text{required probability} = \frac{{}^9C_3 \cdot 3^6}{4^9}$$

$$= \frac{{}^9C_3}{27} \cdot \left(\frac{3}{4}\right)^9$$

$$= \frac{28}{9} \cdot \left(\frac{3}{4}\right)^9 \Rightarrow k = \frac{28}{9}$$

Which satisfies $|x - 3| < 1$

17. Ans (B)



Clearly RS is latus-rectum

$$\therefore VF = 2 = a$$

$$\therefore RS = 4a = 8$$

$$\text{Now } OF = 2a = 4$$

$$\Rightarrow \text{Area of triangle ORS} = 16$$

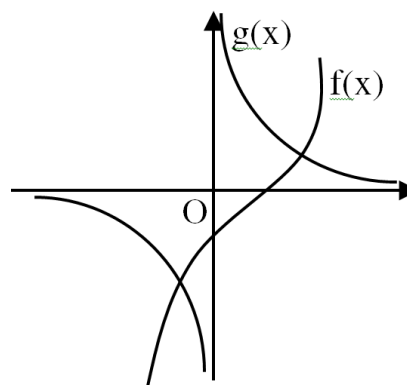
18. Ans (A)

$$e^{6x} - e^{4x} - 2e^{3x} - 12e^{2x} + e^x + 1 = 0$$

$$\Rightarrow (e^{3x} - 1)^2 - e^x (e^{3x} - 1) = 12e^{2x}$$

$$(e^{3x} - 1)^2 (e^x - e^{-x} - e^{-2x}) = 12$$

$$\underbrace{e^x - e^{-x} - e^{-2x}}_{\text{increasing (let } f(x))} = \frac{12}{\underbrace{e^{3x} - 1}_{\text{decreasing (let } g(x))}}$$



$\Rightarrow$ No. of real roots = 2

19. Ans (C)

$$\frac{3}{2a^2} + \frac{1}{b^2} = 1 \quad \text{and} \quad 1 - \frac{b^2}{a^2} = \frac{1}{3}$$

$$\Rightarrow a^2 = 3 \quad b^2 = 3$$

$$\Rightarrow \frac{x^2}{3} + \frac{y^2}{2} = 1 \quad \dots(i)$$

Its focus is (1,0)

Now, eqn. of circle is

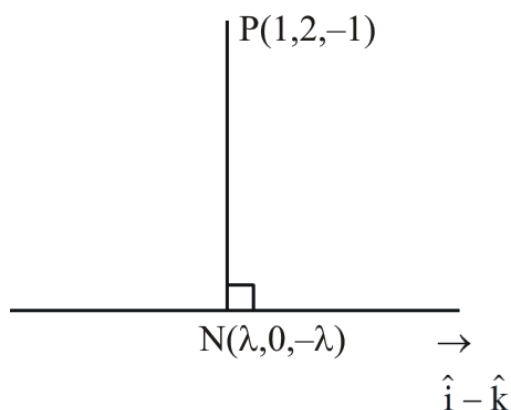
$$(x - 1)^2 + y^2 = \frac{4}{3} \quad \dots(ii)$$

Solving (i) and (ii) we get

$$y = \pm \frac{2}{\sqrt{3}}, x = 1$$

$$\Rightarrow PQ^2 = \left(\frac{4}{\sqrt{3}}\right)^2 = \frac{16}{3}$$

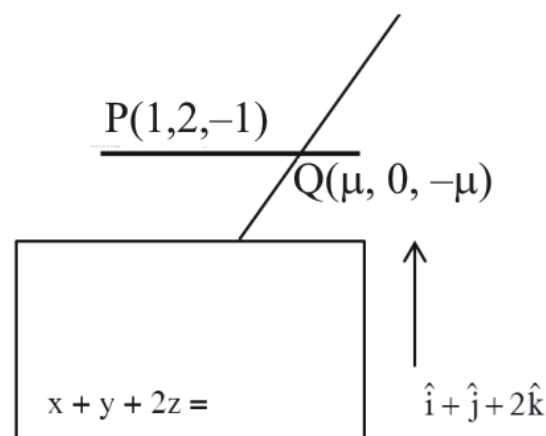
20. Ans (C)



$$\vec{PN} \cdot (\hat{i} - \hat{k}) = 0$$

$$\Rightarrow N(1, 0, -1)$$

Now,



$$\vec{PQ} \cdot (\hat{i} + \hat{j} + 2\hat{k}) = 0$$

$$\Rightarrow \mu = -1$$

$$\Rightarrow Q(-1, 0, 1)$$

$$\vec{PN} = 2\hat{j}$$

and

$$\vec{PQ} = 2\hat{i} + 2\hat{j} - 2\hat{k}$$

$$\Rightarrow \cos \alpha = \frac{1}{\sqrt{3}}$$

PART-3 : MATHEMATICS

SECTION-II

1. Ans (4)

$$\text{Let } e^y = t$$

$$\Rightarrow \frac{dt}{dx} - (2 \sin x)t = -\sin x \cos^2 x$$

$$\text{I.F.} = e^{2 \cos x}$$

$$\Rightarrow t \cdot e^{2 \cos x} = \int e^{2 \cos x} \cdot (-\sin x \cos^2 x) dx$$

$$\Rightarrow e^y \cdot e^{2 \cos x} = \int e^{2z} \cdot z^2 dz, z = e^{2 \cos x}$$

$$\Rightarrow e^y \cdot e^{2 \cos x} = \frac{1}{2} \cdot \cos^2 x \cdot e^{2 \cos x} - \frac{1}{2} \cos x \cdot e^{2 \cos x} + \frac{e^{2 \cos x}}{4} + C$$

$$\text{at } x = \frac{\pi}{2}, y = 0 \Rightarrow C = \frac{3}{4}$$

$$\Rightarrow e^y = \frac{1}{2} \cos^2 x - \frac{1}{2} \cos x + \frac{1}{4} + \frac{3}{4} \cdot e^{-2 \cos x}$$

$$\Rightarrow y = \log \left[\frac{\cos^2 x}{2} - \frac{\cos x}{2} + \frac{1}{4} + \frac{3}{4} e^{-2 \cos x} \right]$$

$$\text{Put } x = 0$$

$$\Rightarrow y = \log \left[\frac{1}{4} + \frac{3}{4} e^{-2} \right] \Rightarrow \alpha = \frac{1}{4}, \beta = \frac{3}{4}$$

2. Ans (3)

$$\ell = \left(\underbrace{1 + \frac{2}{3} + \frac{6}{3^2} + \frac{10}{3^3} + \dots}_S \right)^{\log_{0.25} \left(\frac{1}{3} + \frac{1}{3^2} + \dots \right)}$$

$$S = 1 + \frac{2}{3} + \frac{6}{3^2} + \frac{10}{3^3} + \dots$$

$$\frac{S}{3} = \frac{1}{3} + \frac{2}{3^2} + \frac{6}{3^3} + \dots$$

$$- - - -$$

$$\frac{2S}{3} = 1 + \frac{1}{3} + \frac{4}{3^2} + \frac{4}{3^3} + \dots$$

$$\frac{2S}{3} = \frac{4}{3} + \frac{4}{3^2} + \frac{4}{3^3} + \dots$$

$$S = \frac{3}{2} \left(\frac{4/3}{1 - 1/3} \right) = 3$$

$$\text{Now } \ell = (3)^{\log_{0.25} \left(\frac{1/3}{1 - 1/3} \right)}$$

$$\ell = 3^{\log_{(1/4)} \left(\frac{1}{2} \right)} = 3^{1/2} = \sqrt{3}$$

$$\Rightarrow \ell^2 = 3$$

3. **Ans (164)**

$\therefore$ Sum of frequencies = 584

$$\Rightarrow \alpha + \beta = 390$$

Now, Median is at $\frac{584}{2} = 292^{\text{th}}$

$\therefore$ Median = 45 (lies in class 40 – 50)

$$\Rightarrow \alpha + 110 + 54 + 15 = 292$$

$$\Rightarrow \alpha = 113, \beta = 277$$

$$\Rightarrow |\alpha - \beta| = 164$$

4. **Ans (3)**

$$\vec{p} = 2\hat{i} + 3\hat{j} + \hat{k} \text{ (Given)}$$

$$\vec{q} = \hat{i} + 2\hat{j} + \hat{k}$$

$$\text{Now } (\vec{p} + \vec{q}) \times (\vec{p} - \vec{q}) = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3 & 5 & 2 \\ 1 & 1 & 0 \end{vmatrix}$$

$$= -2\hat{i} - 2\hat{j} - 2\hat{k}$$

$$\Rightarrow \vec{r} = \pm \sqrt{3} \frac{((\vec{p} + \vec{q}) \times (\vec{p} - \vec{q}))}{|(\vec{p} + \vec{q}) \times (\vec{p} - \vec{q})|} = \pm \frac{\sqrt{3}(-2\hat{i} - 2\hat{j} - 2\hat{k})}{\sqrt{2^2 + 2^2 + 2^2}}$$

$$\vec{r} = \pm (-\hat{i} - \hat{j} - \hat{k})$$

According to question

$$\vec{r} = \alpha\hat{i} + \beta\hat{j} + \gamma\hat{k}$$

$$\text{So } |\alpha| = 1, |\beta| = 1, |\gamma| = 1$$

$$\Rightarrow |\alpha| + |\beta| + |\gamma| = 3$$

5. **Ans (1)**

Coeff. of middle term in $(1+x)^{20} = {}^{20}C_{10}$

& Sum of Coeff. of two middle terms in

$$(1+x)^{19} = {}^{19}C_9 + {}^{19}C_{10}$$

$$\text{So required ratio} = \frac{{}^{20}C_{10}}{{}^{19}C_9 + {}^{19}C_{10}} = \frac{{}^{20}C_{10}}{{}^{20}C_{10}} = 1$$

6. **Ans (16)**

$$|A| = ad - bc = 15$$

where $a, b, c, d \in \{\pm 3, \pm 2, \pm 1, 0\}$

Case I $ad = 9$ & $bc = -6$

For ad possible pairs are (3,3), (-3,-3)

For bc possible pairs are (3,-2), (-3,2), (-2,3), (2,-3)

$$\text{So total matrix} = 2 \times 4 = 8$$

Case II $ad = 6$ & $bc = -9$

Similarly total matrix = $2 \times 4 = 8$

$$\Rightarrow \text{Total such matrices are} = 16$$

7. **Ans (238)**

Class	10 th	11 th	12 th
Total student	5	6	8
	2	3	5
	$\Rightarrow {}^5C_2 \times {}^6C_3 \times {}^8C_5$		
Number of selection	2	2	6
	$\Rightarrow {}^5C_2 \times {}^6C_2 \times {}^8C_6$		
	3	2	5
	$\Rightarrow {}^5C_3 \times {}^6C_2 \times {}^8C_5$		

$$\Rightarrow \text{Total number of ways} = 23800$$

According to question

$$100K = 23800$$

$$\Rightarrow K = 238$$

8. **Ans (1)**

$$x^2 + 5\sqrt{2}x + 10 = 0$$

$$\& p_n = \alpha^n - \beta^n \text{ (Given)}$$

Now

$$\frac{P_{17}P_{20} + 5\sqrt{2}P_{17}P_{19}}{P_{18}P_{19} + 5\sqrt{2}P_{18}^2} = \frac{P_{17}(P_{20} + 5\sqrt{2}P_{19})}{P_{18}(P_{19} + 5\sqrt{2}P_{18})}$$

$$\frac{P_{17}(\alpha^{20} - \beta^{20} + 5\sqrt{2}(\alpha^{19} - \beta^{19}))}{P_{18}(\alpha^{19} - \beta^{19} + 5\sqrt{2}(\alpha^{18} - \beta^{18}))} \\ \frac{P_{17}(\alpha^{19}(\alpha + 5\sqrt{2}) - \beta^{19}(\beta + 5\sqrt{2}))}{P_{18}(\alpha^{18}(\alpha + 5\sqrt{2}) - \beta^{18}(\beta + 5\sqrt{2}))}$$

$$\text{Since } \alpha + 5\sqrt{2} = -10/\alpha \quad \dots(1)$$

$$\& \beta + 5\sqrt{2} = -10/\beta \quad \dots(2)$$

Now put there values in above expression

$$= -\frac{10P_{17}P_{18}}{-10P_{18}P_{17}} = 1$$

9. **Ans (210)**

$$\left(\left(x^{1/3} + 1 \right) - \left(\frac{x^{1/2} + 1}{x^{1/2}} \right) \right)^{10} \\ = \left(x^{1/3} - \frac{1}{x^{1/2}} \right)^{10}$$

Now General Term

$$T_{r+1} = {}^{10}C_r \left(x^{1/3} \right)^{10-r} \cdot \left(-\frac{1}{x^{1/2}} \right)^r$$

For independent term

$$\frac{10-r}{3} - \frac{r}{2} = 0 \Rightarrow r = 4$$

$$\Rightarrow T_5 = {}^{10}C_4 = 210$$

10. Ans (11)

$$\text{Let } X = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \text{ \& } A = \begin{pmatrix} 0 & i \\ 1 & 0 \end{pmatrix}^n$$

$$\Rightarrow AX = IX$$

$$\Rightarrow A = I$$

$$\Rightarrow \begin{pmatrix} 0 & i \\ 1 & 0 \end{pmatrix}^n = I$$

$$\Rightarrow A^8 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$\Rightarrow n$ is multiple of 8

So number of 2 digit numbers in the set

$$S = 11 \text{ (16, 24, 32, ,96)}$$